

CredResolve Production Analytics Architecture

Production design for controlled collection analytics, governed metrics, analytical investigation and executive decision-making

1. Architecture Objective

The CredResolve production analytics architecture converts operational collections data into validated analytical information and finally into business decisions.

The architecture prevents raw operational inconsistencies from directly affecting recovery metrics or investment decisions.

Core principle: Raw operational data should not directly drive business decisions. Data must pass through controlled validation, cleaning, standardization and analytical layers first.

2. Architecture Principles

Principle	Implementation
Source preservation	Raw operational records are preserved before transformation.
Controlled transformation	Ingestion, validation, cleaning and business logic are separated.
Account-centric analysis	account_id is the canonical analytical entity for recovery analysis.
Governed metrics	Metrics use documented definitions and appropriate denominators.
Reproducibility	SQL, Python and Golden-layer transformations are reproducible.
Traceability	Analytical results can be traced back through the data layers.
Production controls	Quality, governance, monitoring and orchestration operate across the pipeline.

3. Business Objectives Supported

- Reconstruct monthly recovery and collection performance.
- Identify meaningful changes in activity and performance.
- Investigate portfolio, DPD, risk, campaign, channel, agent and vendor effects.
- Validate payment attribution and analytical entity relationships.
- Evaluate the reported 11% recovery improvement.
- Perform observational counterfactual analysis.
- Evaluate the proposed ■10 Cr investment.
- Produce governed executive reporting.

4. Data Architecture Layers

Data Sources

Operational collections datasets including accounts, borrowers, payments, calls, campaigns, agents, vendors and collection events.

Acquire

Batch and incremental ingestion brings source data into the analytical environment while preserving source context.

Raw / Landing

The source-preservation layer used for traceability, auditability and reprocessing. Business metrics are not calculated directly here.

Staging

Structural preparation including schema normalization, datatype handling, identifier normalization and timestamp parsing.

Quality Gate

Validation of identifiers, duplicates, relationships, timestamps, payment attribution and other structural quality conditions.

Clean Layer

Validated and standardized analytical datasets. The current validation framework recorded 49 of 49 checks passed.

Golden Layer

Stable analytical representation derived from the validated Clean layer. It provides the controlled source for downstream analytical work.

Golden Layer Analytical Decisions

Dataset	Analytical role
Accounts	account_id is the canonical recovery-analysis entity.
Payments	Payment attribution follows payment_id → account_id.
Calls	Call activity connects accounts, agents, campaigns and vendors.
Campaigns	campaign_id provides campaign and strategy context.
Agents	agent_id provides operational identity and tenure context.
Collection events	Supporting event-level evidence for contact and recovery analysis.

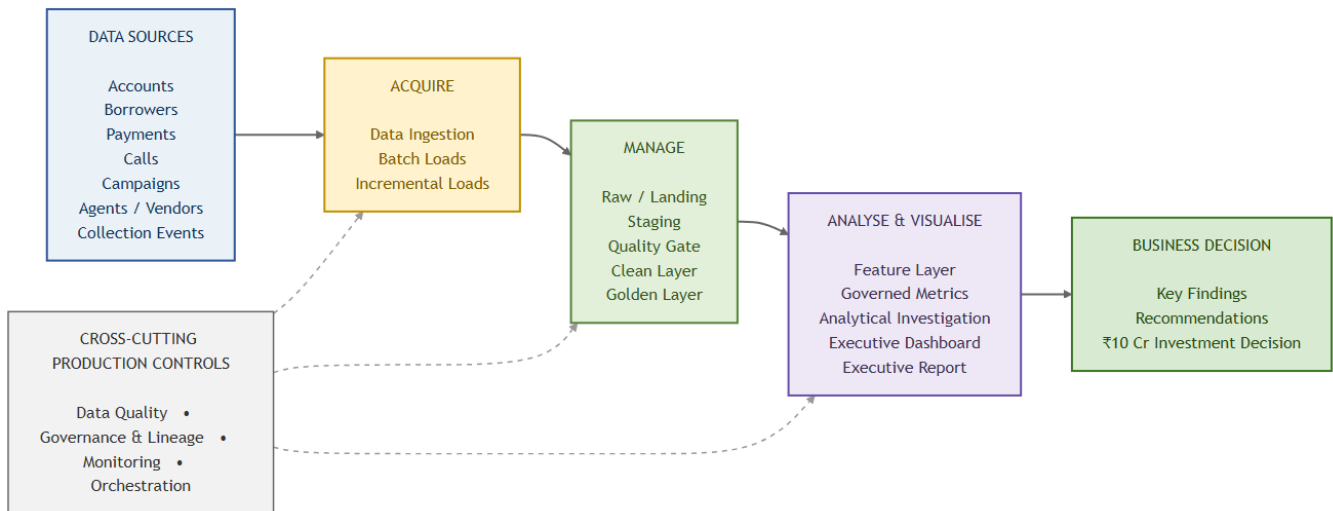
5. Data Quality Boundary

Data-quality issues are not silently converted into business assumptions. Records that remain analytically usable can be retained with explicit quality indicators, while ambiguous issues remain visible for investigation.

6. Architecture Overview

The main production flow is intentionally simple:

Data Sources → Acquire → Manage → Analyse & Visualise → Business Decision



The architecture separates the main analytical pipeline from the cross-cutting production controls. Data Quality, Governance & Lineage, Monitoring and Orchestration support the pipeline without creating unnecessary dependencies between analytical stages.

The Golden layer acts as the controlled hand-off between data management and analytical processing. Features and governed metrics are produced from this stable analytical foundation.

7. Analytical Processing

Layer	Purpose
Feature Layer	Reusable DPD, risk, status, loan, agent, campaign, channel and contact features.
Governed Metrics	Standardized recovery, payment, contact and conversion measurements.
Analytical Investigation	Forensic, statistical, driver, counterfactual and improvement analysis.
Executive Outputs	Dashboard, executive memo and decision-support reporting.
Business Decision	Recommendations and investment decisions based on governed evidence.

8. Production Operations and Governance

Governance and Lineage

Every analytical output should be traceable through Source → Raw/Landing → Staging → Quality Gate → Clean → Golden → Feature → Metric → Analysis → Output.

Incremental Processing

New or changed records should be processed without rebuilding unaffected historical data. Downstream features and metrics affected by those records should be refreshed.

Late-Arriving Data

Event timestamp and ingestion timestamp should be retained separately. Late records should trigger recalculation of affected reporting periods.

Backfills

Historical corrections and transformation changes should use reproducible backfill processes rather than manual modification of analytical results.

Monitoring

Monitor volume, missing identifiers, duplicates, schema changes, validation failures, late records, recovery anomalies and portfolio-mix changes.

Anomaly Detection

Unexpected movements in payment volume, recovered amount, answer rate, attempt frequency or portfolio composition should be flagged for investigation.

Metric Governance

Every metric requires an explicit numerator, denominator, population, time window and attribution definition.

9. Production Operating Model

Stage	Primary responsibility
Source	Operational collection records
Acquire	Batch and incremental acquisition
Raw / Landing	Source preservation and traceability
Staging	Structural preparation
Quality Gate	Validation and exception identification
Clean	Validated analytical datasets
Golden	Stable analytical source
Features	Reusable analytical attributes
Metrics	Governed business measurements
Analysis	Evidence and decision support
Outputs	Dashboard and executive reporting
Decision	Recommendation and investment action

Production principle: A business decision is the final output of a controlled analytical chain, not a direct interpretation of raw operational data.

Architecture status: designed for reproducible analytical processing, governed metrics, controlled investigation and executive decision support.